

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0575

Mill C 345 kV
March 24, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0575. The Interconnection Customer proposes a 1028.6 MW natural gas combined cycle facility interconnecting at the Mill C 345 kV substation in the New Jersey area. The request was received on January 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 290 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,235,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Allston – Metuchen 138kV: 124.4% loading (256 MVA) following the outage of the Allston – Roy C Miller 345kV Line; DFAX = 5.4%.

For the loss of Allston – Roy C Miller 345kV L, the Lafayette Rd Sta – Metuchen Substation 138kV L loads to 116.2% of its 245 MVA rating. Project DFAX: 5.4%.

For the loss of Roy C Miller – Allston 345kV L, the Lafayette Road Station – Secod 138kV loads to 108.0% of its 230 MVA rating. Project DFAX: 5.4%.

For the loss of Laurel Ave – Crawfordsville Power Plant Substation 345kV, the Pine Tree Acres – Crawfordsville Power Plant 138kV line loads to 101.1% of its 432 MVA rating. Project DFAX: 5.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Allston – Metuchen 138kV line	Rebuild line with higher-capacity conductor	\$4,312,000	81% (\$3,477,000)
Lafayette Rd Sta – Metuchen Substation 138kV L	Rebuild line with higher-capacity conductor	\$1,064,000	81% (\$858,000)
Lafayette Rd Sta – Secod 138kV Line	Reconductor line	\$926,000	81% (\$747,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pine Tree Acres – Crawfordsville Power Plant 138kV line	Reconductor line	\$308,000	50% (\$153,000)
	Total allocated to project		\$5,235,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.